

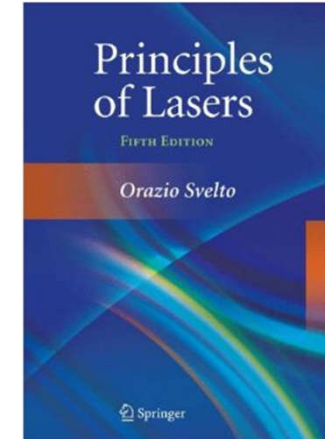


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch4-3 Oscillation Characteristics of Lasers



Dr. Dan Luo(罗丹)

Email: luod@sustech.edu.cn

Office: Room 331 at College
of Engineering South Tower

Tel: 0755-88018552



Chapter 4: Oscillation Characteristics

❖ 4.4 Relaxation Oscillation in Pulse Lasers

- Qualitative interpretation of relaxation oscillation
- Theoretical treatment of relaxation oscillation
- Practical significance of relaxation oscillation

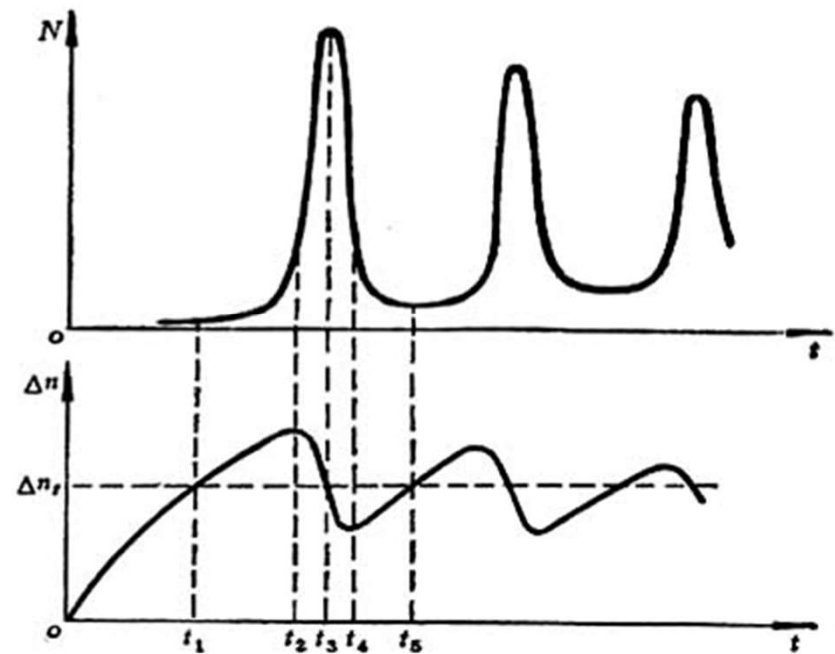
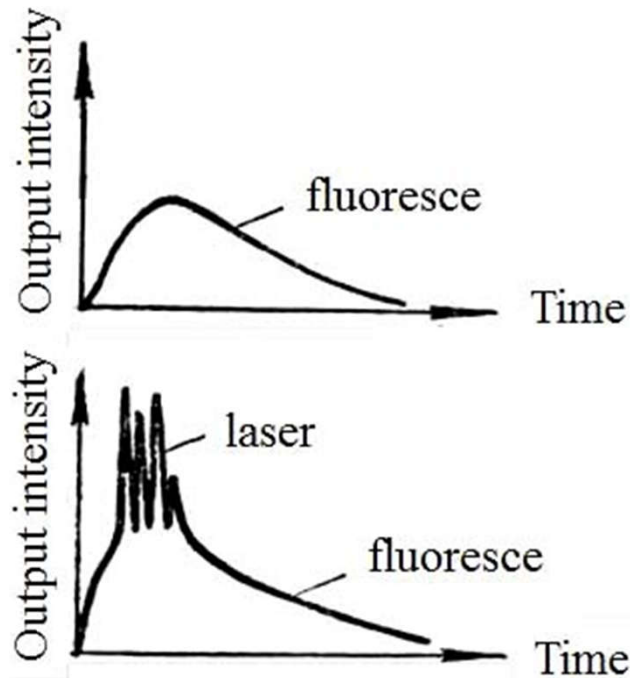
❖ 4.5 Line-Width Limit of Single Mode Lasers

- Passive cavity line-width limit
- Active cavity line-width limit

❖ 4.6 Frequency Pulling of Laser

4.4 Relaxation Oscillation in Pulse Lasers

- ❖ Qualitative interpretation of the formation of relaxation oscillations (transient behaviors of Δn and N_l)



➤ t_1-t_2 The increase rate of Δn
due to pumping

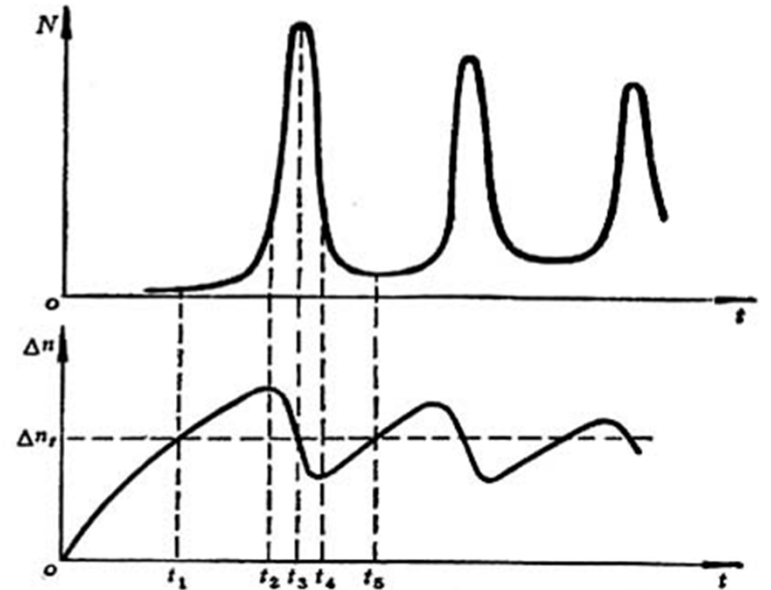
> The decrease rate of Δn
due to stimulated emission

$$\Delta n \uparrow \rightarrow \Delta n = \Delta n_t (t = t_1) \rightarrow N_l \uparrow, \Delta n \uparrow (t_1 < t < t_2)$$

- t_2 - t_3 Stimulated radiation causes N_l to rise sharply $\rightarrow \Delta n \downarrow$

$$\frac{dN_l}{dt} = -\frac{d\Delta n}{dt}(t = t_2) \rightarrow \begin{matrix} N_l \uparrow \\ \Delta n \downarrow \end{matrix} (t > t_2)$$

$$\rightarrow \frac{N_l}{\Delta n_t}(t = t_3) \text{ maximum}$$



- t_3 - t_4 Stimulated radiation causes $\Delta n < \Delta n_t$, $N_l \downarrow$ sharply, $\Delta n \downarrow$

- t_4 - t_5 { If pump hasn't stopped yet, weakening of stimulated radiation leads to

$$N_l \downarrow \rightarrow \Delta n \uparrow \rightarrow \Delta n = \Delta n_t$$

 Rate of Δn reduction by stimulated radiation = Rate of Δn increase by pump

- $t > t_5$ Start the evolution of 2nd pulse. This process occurs repeatedly during the pumping process. As a result, the laser shows a series of spikes.

- The relaxation oscillation process occurs during the establishment of a laser. During pumping, both the inverted population and photon density undergo rapid changes due to STE. As a result, the output exhibits the characteristics of relaxation oscillations, forming multiple spikes.
- The stronger the pump, the higher the damping oscillation frequency, leading to a faster decay.

- ❖ Theoretical treatment of the relaxation oscillation process (solve the rate equation at non-steady-state)
 - 1) Exact solution: numerical method
 - ✓ 2) Approximate solution: 1st order perturbation of steady-state

Qualitative interpretation is OK.

The quantitative comparison does not align well with the experimental results. This discrepancy arises because Δn and N are not merely small fluctuations around the equilibrium state; instead, they may exhibit large fluctuations or intermittent oscillations..

Example: Four-level system, single-mode $\nu = \nu_0$, $n_2 \approx \Delta n$

Assume $\eta_F = \eta_1 = \eta_2 = 1$ $L = l$ $\eta_1 = \frac{S_{32}}{S_{32} + A_{30}}$ $\eta_2 = \frac{A_{21}}{A_{21} + S_{21}}$

$$\left[\begin{array}{l} \frac{dN(t)}{dt} = \Delta n(t) \sigma_{21} \nu N(t) - \frac{N(t)}{\tau_R} \\ \frac{d\Delta n(t)}{dt} = -\Delta n(t) \sigma_{21} \nu N(t) - \Delta n(t) A_{21} + [n - \Delta n(t)] W_{03} \end{array} \right.$$

In the 1st order perturbation approximation, assume that the transient photon numbers and the inverted populations fluctuate slightly near their corresponding equilibrium values.

$$\begin{array}{ll} N(t) = N_0 + N'(t) & N'(t) \ll N_0 \\ \Delta n(t) = (\Delta n)_0 + \Delta n'(t) & \Delta n'(t) \ll (\Delta n)_0 \end{array}$$

N_0 and $(\Delta n)_0$ steady-state solution

$N'(t)$ and $\Delta n'(t)$ small value

(1) Steady-state ($dX/dt=0$): N_0 and $(\Delta n)_0$

$$\frac{dN(t)}{dt} = \Delta n(t) \sigma_{21} \nu N(t) - \frac{N(t)}{\tau_R} \Rightarrow \Delta n(t) = \frac{1}{\tau_R \sigma_{21} \nu} = \frac{\delta}{\sigma_{21} l} = \Delta n_t \quad (\Delta n)_0 = \Delta n_t$$

$$\frac{d\Delta n(t)}{dt} = -\Delta n(t) \sigma_{21} \nu N(t) - \Delta n(t) A_{21} + [n - \Delta n(t)] W_{03}$$

$$N_0 = \frac{1}{\Delta n_t \sigma_{21} \nu} [W_{03} (n - \Delta n_t) - A_{21} \Delta n_t] = \tau_R [W_{03} (n - \Delta n_t) - A_{21} \Delta n_t]$$

(2) Approximate solution with 1st order perturbation

$$\begin{aligned} \frac{dN'}{dt} &= \left\{ [\Delta n_t + \Delta n'(t)] \sigma_{21} \nu [N_0 + N'(t)] \right\} - \left[\frac{N_0 + N'(t)}{\tau_R} \right] \\ &= \sigma_{21} \nu \left[\cancel{\Delta n_t N_0} + \cancel{\Delta n_t N'(t)} + \Delta n'(t) N_0 + \Delta n'(t) N'(t) \right] - \frac{\cancel{N_0}}{\tau_R} - \frac{\cancel{N'(t)}}{\tau_R} = \sigma_{21} \nu N_0 \Delta n' \end{aligned}$$

$$\frac{d\Delta n'}{dt} = -(\underbrace{\sigma_{21} \nu N_0 + A_{21} + W_{03}}_{=\alpha}) \Delta n' - \frac{1}{\tau_R} N' \quad \tau_R = \beta$$

$$\Rightarrow \begin{aligned} \frac{d^2 \Delta n'}{dt^2} + \alpha \frac{d \Delta n'}{dt} + \beta \gamma \Delta n' &= 0 \\ \frac{d^2 N'}{dt^2} + \alpha \frac{d N'}{dt} + \beta \gamma N' &= 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} \Delta n'(t) &= \Delta n'(0) e^{-\varphi t} \sin \Omega_R t \\ N'(t) &= N'(0) e^{-\varphi t} \sin \left(\Omega_R t - \frac{\pi}{2} \right) \end{aligned}$$

$t=0 \quad \Leftrightarrow \Delta n \rightarrow \Delta n_t$

- $t > 0$ $\Delta n'(t)$, $\Delta N'(t)$ show damped oscillation with the same damping period

Attenuation coefficient

$$\varphi = \frac{\alpha}{2} = \frac{1}{2} (W_{03} + A_{21} + \sigma_{21} \nu N_0)$$

Damped oscillation frequency

$$\Omega_R = \frac{1}{2} \sqrt{4\beta\gamma - \alpha^2} = \sqrt{\beta\gamma - \varphi^2}$$

$$\xrightarrow[n \gg \Delta n_t, 1/\tau_R \gg W_{03}]{A_{21} \Delta n_t \approx (W_{03})_t n}$$

$$\begin{aligned} \varphi &= \frac{1}{2} \sigma_{21} \nu \tau_R W_{03} n \\ \Omega_R &\approx \sqrt{\frac{A_{21}}{\tau_R} \left[\frac{W_{03}}{(W_{03})_t} - 1 \right]} \end{aligned}$$

- $W_{03} \uparrow \rightarrow \varphi \uparrow, \Omega_R \uparrow$

- Multi-mode: For each mode, amplitude, φ and Ω_R are different. Therefore, random superposition and random fluctuation of intensity exist.

$$\Delta n'(t) = \Delta n'(0) e^{-\varphi t} \sin \Omega_R t$$

$$N'(t) = N'(0) e^{-\varphi t} \sin \left(\Omega_R t - \frac{\pi}{2} \right)$$

$t \gg 1/\varphi$ $\Delta n'(t), N'(t) \rightarrow 0$ $N'(t) \rightarrow N_0$, $\Delta n'(t) \rightarrow (\Delta n)_0 = \Delta n_t \Rightarrow$ steady state

❖ Practical significance of relaxation oscillation (pros and cons)

- The relaxation oscillation reflects the instability of the laser oscillation process. It is universal. The relaxation oscillation is the effect of laser oscillation on the transition from a non-steady-state to a steady-state.
- Relaxation oscillation in CW lasers will lead to instability of laser output, which is called noise.
- ✓ warm-up time
- ✓ suddenly change the pump and loss will lead to oscillation instability

4.5 Line-width Limit of Single Mode Lasers

monochromatic $\sim$ temporal coherence

- ❖ Passive cavity line-width: line-width of the eigen cavity mode $\sim$ photon lifetime

Passive cavity ($g=0$: equal number of photons absorbed and emitted)
Intensity variation of light propagating in a passive cavity

$$I(t) = I_0 e^{-t/\tau_R} \quad \tau_R = \frac{L'}{c\delta}$$

$$\Rightarrow \Delta\nu_c = \frac{1}{2\pi\tau_R} = \frac{c\delta}{2\pi L'}$$

Example: He-Ne laser, $L=30$ cm, $r_1=1$, $r_2=0.98$, $a\approx 0$, $\eta=1$.

$$\tau_R = 10^{-7} \text{ s}, \quad \Delta\nu_c = 1.6 \times 10^6 \text{ s}^{-1}$$

❖ Active cavity linewidth ($g > 0$)

$$\delta_s = \delta - g(\nu, I_\nu)l \Rightarrow \Delta\nu_s = \frac{c\delta_s}{2\pi L'} \Rightarrow \boxed{\Delta\nu_s < \Delta\nu_c} \quad \tau'_R = L'/(c\delta_s)$$

Starting from the photon lifetime $\Delta\nu_s \rightarrow 0$? | linewidth of the active cavity is smaller than passive one

- Spontaneous emission always exists in the process of oscillation, laser linewidth can NOT be 0

Stimulated Emission	+	Spontaneous Emission	=	Loss	$P_0 = P_{sp} + P_{ste}$ $gl < \delta \quad \delta_s \neq 0$
Coherent		Noncoherent			

- ✓ Existence of spontaneous emission → loss is not zero in active cavity
- ✓ Noncoherent spontaneous emission → random phase → a **finite and slightly attenuated wave train** of coherent radiation
- ✓ The line-width of the active cavity is determined by the net loss

❖ Estimation of $\Delta \nu_s$ (Four-level system, single mode, $L=l$) $\Delta \nu_s = \frac{c\delta_s}{2\pi L'}$

1. Get δ_s

$$\frac{dN_l}{dt} = \underbrace{\Delta n \sigma_{21}(\nu, \nu_0) \nu N_l}_{\equiv g(\nu, I_\nu)} + \boxed{a_l n_2} - \frac{N_l}{\tau_R}$$

$$a_l = \frac{A_{21} \tilde{g}(\nu, \nu_0)}{n_\nu SL} = \frac{\sigma_{21}(\nu, \nu_0) \nu}{SL} \quad \text{SP probability distributed to the mode } l$$

stable oscillation

$$g(\nu, I_\nu) \nu N_l + a_l n_2 - \frac{N_l \delta \nu}{L} = 0 \quad n_2 = n_{2t} \quad \Delta n = \Delta n_t$$

$$\delta_s = \delta - g(\nu, I_\nu) l = \boxed{a_l n_{2t}} \frac{L}{\nu N_l}$$

net loss $\neq 0$
compensated by SP

2. Get a_l, N_l

$$\begin{aligned} P_{out} &= P_{sp} + P_{ste} \approx P_{ste} \\ &= \Delta n_t \sigma_{21}(\nu, \nu_0) \nu N_l SL h\nu \\ &= \Delta n_t a_l N_l (SL)^2 h\nu \end{aligned} \quad \Rightarrow \quad \begin{aligned} P_{out} &= \frac{N_l}{2} \nu Sh\nu T \\ \Downarrow \\ N_l &= \frac{2P_{out}}{T\nu Sh\nu} \Rightarrow a_l = \frac{T\nu}{2SL^2 \Delta n_t} \end{aligned}$$

3. Theoretical line-width limit of a single mode laser

insert N_l, a_l into δ_s
insert δ_s

$$\Delta \nu_s = \frac{n_{2t}}{\Delta n_t} \frac{2\pi (\Delta \nu_c)^2 h\nu}{P_{out}} \approx \frac{n_{2t}}{\Delta n_t} \frac{2\pi (\Delta \nu_c)^2 h\nu_0}{P_{out}}$$

Example: He-Ne, $\lambda=632.8$ nm, $L=30$ cm, $T=0.02$, $P_{out}=1$ mW

$$\Delta \nu_c = 1.6 \times 10^6 \text{ Hz}, \Delta \nu_s = 6 \times 10^{-3} \text{ Hz}, n_{2t}/\Delta n_t = 1$$

❖ The line-width of the real lasers

➤ Many orders of magnitude far beyond theoretical line-width limit

Reason: Frequency drift caused by temperature and other factors

- ✓ He-Ne laser (glass shell) 0.01 °C $\rightarrow$ 1 MHz
- ✓ Solid-state laser $\Delta T \rightarrow \nu_0$ shift

Summary $\Delta \nu_c = \frac{1}{2\pi\tau_R} = \frac{c\delta}{2\pi L'}$ $\Delta \nu_s = \frac{c\delta_s}{2\pi L'} = \frac{ca_l n_{2t} L / \nu N_l}{2\pi L'} = \frac{n_{2t}}{\Delta n_t} \frac{2\pi (\Delta \nu_c)^2 h\nu_0}{P_{out}}$

- Physical mechanism of line-width (single mode laser): SP, noise
- $\Delta \nu_s \ll \Delta \nu_c$
- $P_{out} \uparrow \rightarrow \Delta \nu_s \downarrow$ intra-cavity coherent photon number $\uparrow$, STE accounts for a larger proportion of P_{out}

$$\bar{n} = \frac{\rho_\nu}{8\pi h\nu^3 / c^3} = \frac{B_{21}}{A_{21}} \rho_\nu = \frac{W_{21}}{A_{21}} \quad \text{for He-Ne: } \bar{n} \approx 3 \times 10^8$$

$$\bar{n} = \frac{\text{Total photon number}}{\text{Mode number}} \Rightarrow \frac{W_{21}}{\text{Total photon number}} = \frac{A_{21}}{\text{Mode number}}$$

- $\Delta \nu_s \propto (\Delta \nu_c)^2$
 - ✓ $\delta \downarrow \rightarrow \Delta \nu_c \downarrow \rightarrow \Delta \nu_s \downarrow$
 - ✓ $L \uparrow \rightarrow \Delta \nu_c \downarrow \rightarrow \Delta \nu_s \downarrow$ vertical external cavity semiconductor laser
- The actual line-width of a single-mode laser highly depends on the frequency stability

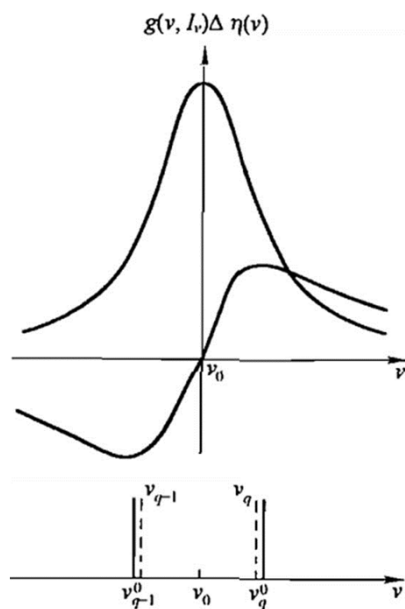
4.6 Frequency Pulling of Laser

➤ What is frequency pulling?

➤ Physical mechanism: dispersion

➤ Dispersion induced frequency pulling $\Delta\eta(\nu) = \frac{c(\nu - \nu_0)}{2\pi\nu\Delta\nu_H} g_H(\nu, I_\nu)$

$$\eta = \eta_0 \left[1 - \frac{(\nu_0 - \nu)c}{\Delta\nu_H \omega} g \right]$$



Passive cavity
longitudinal frequency

$$\nu_q^0 = q \frac{c}{2\eta^0 L}$$

Active cavity
longitudinal frequency

$$\nu_q = q \frac{c}{2\eta L} = q \frac{c}{2(\eta^0 + \Delta\eta)L}$$

Frequency difference
between active and
passive cavity

$$\begin{aligned} \nu_q - \nu_q^0 &= q \frac{c}{2(\eta^0 + \Delta\eta)L} - q \frac{c}{2\eta^0 L} \\ &\approx \frac{-\Delta\eta}{\eta^0} \nu_q^0 \end{aligned}$$

In real lasers, the dispersion of gain material makes longitudinal frequency approach more closer to the central frequency compared to the passive cavity:

Frequency Pulling

- Discuss the refractive index change and pulling parameter (homogeneous and inhomogeneous broadening)

homogeneous broadening

$$\Delta\eta(\nu) = \frac{c(\nu - \nu_0)}{2\pi\nu\Delta\nu_H} g_H(\nu, I_\nu)$$

$$g_H(\nu, I_\nu) = \frac{A_{21}\nu\Delta n^0}{4\pi^2\nu_0^2\Delta\nu_H} \frac{(\Delta\nu_H/2)^2}{(\nu - \nu_0)^2 + (\Delta\nu_H/2)^2 (1 + I_\nu/I_s)}$$

frequency difference between active and passive cavity

$$\nu_q - \nu_q^0 \approx \frac{-\Delta\eta(\nu_q)}{\eta^0} \nu_q^0 \quad \xrightarrow{\nu_q \approx \nu_q^0} \quad \nu_q - \nu_q^0 \approx \frac{-c(\nu_q - \nu_0)}{2\pi\eta^0\Delta\nu_H} g_H(\nu_q, I_{\nu_q})$$

steady-state

$$\nu_q - \nu_q^0 = -\frac{\Delta\nu_c}{\Delta\nu_H} (\nu_q - \nu_0) \quad \xleftarrow{\Delta\nu_c = \frac{c\delta}{2\pi\eta^0 L}} \quad \nu_q - \nu_q^0 = \frac{-c(\nu_q - \nu_0)}{2\pi\eta^0\Delta\nu_H} \frac{\delta}{L}$$

$$\sigma_H = -\frac{\nu_q - \nu_q^0}{\nu_q - \nu_0} = \frac{\Delta\nu_c}{\Delta\nu_H}$$

pulling parameter of
homogeneously broadened laser

**inhomogeneous
broadening**

$$\Delta\eta_i(\nu) = \frac{c}{2\pi^{3/2}\nu} g_i^0(\nu_0) e^{-4\ln 2 \left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2} \int_0^{2\sqrt{\ln 2} \frac{\nu-\nu_0}{\Delta\nu_D}} e^{-t^2} dt$$

$$\Delta\eta_i(\nu) = \frac{c(\nu-\nu_0)}{\pi^{3/2}\nu} \sqrt{\ln 2} g_i^0(\nu_0) e^{-4\ln 2 \left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2}$$

$$\begin{aligned} \nu_q - \nu_q^0 &= -\frac{c(\nu_q - \nu_0)}{\pi^{3/2}\eta^0 \Delta\nu_D} \sqrt{\ln 2} g_i^0(\nu_0) e^{-4\ln 2 \left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2} \\ &= -\frac{c(\nu_q - \nu_0)}{\pi^{3/2}\eta^0 \Delta\nu_D} \sqrt{\ln 2} g_i(\nu_q, I_{\nu_q}) \sqrt{1 + \frac{I_{\nu_q}}{I_s}} \end{aligned}$$

Steady state $g_i(\nu_q, I_{\nu_q}) = \frac{\delta}{L}$ $\rightarrow$ $= -2\sqrt{\frac{\ln 2}{\pi}} \frac{\Delta\nu_c}{\Delta\nu_D} \sqrt{1 + \frac{I_{\nu_q}}{I_s}} (\nu_q - \nu_0)$ $\because \Delta\nu_c = \frac{c\delta}{2\pi\eta^0 L}$

$$\sigma_i = -\frac{\nu_q - \nu_q^0}{\nu_q - \nu_0} = 2\sqrt{\frac{\ln 2}{\pi}} \frac{\Delta\nu_c}{\Delta\nu_D} \sqrt{1 + \frac{I_{\nu_q}}{I_s}}$$

He-Ne laser
 $\sigma_i = 10^{-3}$